

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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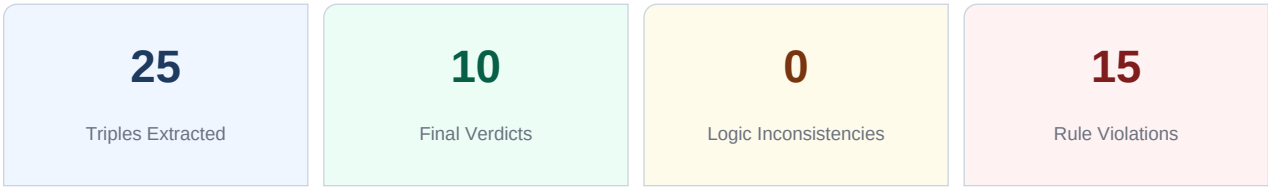
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REPORT OVERVIEW & METHODOLOGY

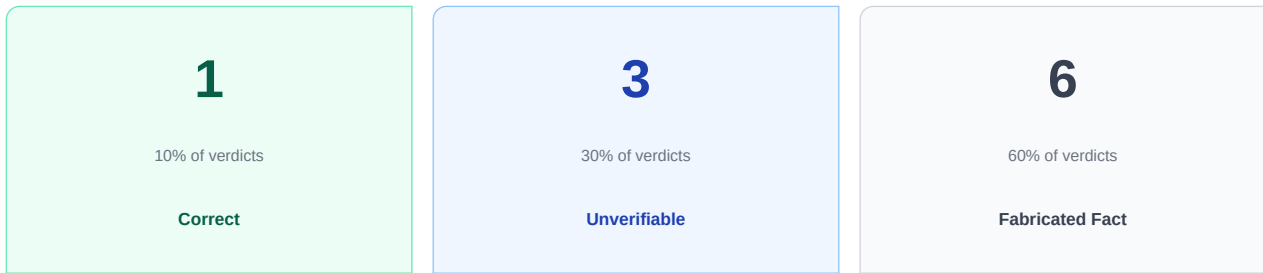
This document outlines the results of the KGVS verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	890	HAS_INTEREST_RATE	30		The interest rate associated with the entity.
2	290	HAS_INTEREST_RATE	20		The interest rate associated with the entity.
3	333	HAS_INTEREST_RATE	40		The interest rate associated with the entity.
4	333	HAS_RISK_TIER	high prime		The risk category assigned to the entity.
5	290	HAS_RISK_TIER	high prime		The risk category assigned to the entity.
6	301	HAS_RISK_TIER	deep subprime		The risk category assigned to the entity.
7	333	HAS_APPROVAL_ODDS	200		The likelihood or score indicating approval chances.
8	290	HAS_APPROVAL_ODDS	30		The likelihood or score indicating approval chances.
9	890	HAS_APPROVAL_ODDS	40		The likelihood or score indicating approval chances.
10	290	HAS_COLLATERAL_TYPE	vehicle		The type of asset pledged as collateral.
11	500	HAS_COLLATERAL_TYPE	vehicle		The type of asset pledged as collateral.
12	600	HAS_COLLATERAL_TYPE	assets		The type of asset pledged as collateral.
13	301	HAS_INTEREST_RATE	2		The interest rate associated with the entity.
14	303	HAS_INTEREST_RATE	3		The interest rate associated with the entity.
15	333	HAS_INTEREST_RATE	4		The interest rate associated with the entity.
16	777	HAS_INTEREST_RATE	6		The interest rate associated with the entity.
17	433	HAS_INTEREST_RATE	27		The interest rate associated with the entity.
18	533	HAS_INTEREST_RATE	21		The interest rate associated with the entity.
19	633	HAS_INTEREST_RATE	32		The interest rate associated with the entity.
20	loan application	REQUIRES_NUMBER_OF_FAMILY_MEMBERS	Not specified in the given context		The number of family members required for the application.
21	300	HAS_DESCRIPTION	smallest credit score		A textual description of the entity.
22	loan	HAS_MINIMUM_AMOUNT_USD	1000		The minimum loan amount in US Dollars.
23	loan	HAS_MAXIMUM_AMOUNT_USD	10000500		The maximum loan amount in US Dollars.
24	305	HAS_APPROVAL_ODDS	1		The likelihood or score indicating approval chances.
25	310	HAS_DEFAULT_PROBABILITY	2		The probability or score indicating the likelihood of default.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	HAS_INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
290	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	40	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
290	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
301	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
333	HAS_APPROVAL_ODDS	200	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_APPROVAL_ODDS	30	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
890	HAS_APPROVAL_ODDS	40	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_COLLATERAL_TYPE	vehicle	HAS_COLLATERAL_TYPE	HIGH
500	HAS_COLLATERAL_TYPE	vehicle	HAS_COLLATERAL_TYPE	HIGH
600	HAS_COLLATERAL_TYPE	assets	HAS_COLLATERAL_TYPE	HIGH
301	HAS_INTEREST_RATE	2	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
303	HAS_INTEREST_RATE	3	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	4	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
777	HAS_INTEREST_RATE	6	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
433	HAS_INTEREST_RATE	27	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
533	HAS_INTEREST_RATE	21	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
633	HAS_INTEREST_RATE	32	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
loan application	REQUIRES_NUMBER_OF_FAMILY_MEMBERS	Not specified in the given context	REQUIRES_NUMBER_OF_FAMILY_MEMBERS	HIGH
300	HAS_DESCRIPTION	smallest credit score	HAS_DESCRIPTION	LOW
loan	HAS_MINIMUM_AMOUNT_USD	1000	HAS_MINIMUM_AMOUNT_USD	LOW
loan	HAS_MAXIMUM_AMOUNT_USD	10000500	HAS_MAXIMUM_AMOUNT_USD	LOW
305	HAS_APPROVAL_ODDS	1	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
310	HAS_DEFAULT_PROBABILITY	2	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	MEDIUM

FACTUALLY INCONSISTENT TRIPLES

All triples labelled `factually_inconsistent` from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	<code>c_interest_rate_range</code>	The subject credit score 890 violates the constraint because it is greater than 850.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	290	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	<code>c_interest_rate_range</code>	The subject credit score 290 violates the constraint because it is less than 300.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	40	<code>c_interest_rate_range</code>	The object interest rate 40 violates the constraint because it is greater than 30.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	333	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	<code>c_tier_300_579</code>	The object risk tier 'high prime' does not semantically contain 'deep subprime' for the subject credit score 333.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	290	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	<code>c_tier_range</code>	The subject credit score 290 violates the constraint because it is less than 300.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	333	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	200	<code>c_approval_odds_range</code>	The object approval odds 200 violates the constraint because it is greater than 100.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	290	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	30	<code>c_approval_odds_range</code>	The subject credit score 290 violates the constraint because it is less than 300.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	890	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	40	<code>c_approval_odds_range</code>	The subject credit score 890 violates the constraint because it is greater than 850.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	290	HAS_COLLATERAL_TYPE	vehicle	<code>c_collateral_type</code>	The subject credit score 290 violates the constraint because it is less than 300.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	500	HAS_COLLATERAL_TYPE	vehicle	<code>c_collateral_type</code>	Subject 500 is outside the valid range of 300-850.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	600	HAS_COLLATERAL_TYPE	assets	<code>c_collateral_type</code>	Subject 600 is outside the valid range of 300-850.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	301	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	2	<code>c_interest_rate_range</code>	Subject 301 is outside the valid credit score range of 300-850.

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	633	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	32	<i>c_interest_rate_range</i>	Object 32 is outside the valid interest rate range of 0-30.
factually_inconsistent	RULE_VIOLATION	305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	1	<i>c1_approval_odds</i>	Subject 305 requires the object to semantically contain 8, but it is 1.
factually_inconsistent	RULE_VIOLATION	310	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	2	<i>c_default_prob_range</i>	The subject (credit score) 310 is outside the valid range of 300-850.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
301	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>STRICT</i>	SUBJECT	●●●●●■ ■■■ 0.60	Subject '301' does not exist in DB candidates for the given predicate and object. <i>Ev: subject: 300, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 305, predicate: RISK_TIER_CLASSIFIED_AS_B...</i>
303	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	3	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>NONE</i>	—	●●●●●■ ■■■ 0.20	No DB candidates exist for the predicate, and the search strategy is null with predicate_known=true.
333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>NONE</i>	—	●●●●●■ ■■■ 0.20	No DB candidates exist for the predicate, and the search strategy is null with predicate_known=true.
777	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	6	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>NONE</i>	—	●●●●●■ ■■■ 0.20	No DB candidates exist for the predicate, and the search strategy is null with predicate_known=true.
433	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	27	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>NONE</i>	—	●●●●●■ ■■■ 0.20	No DB candidates exist for the predicate, and the search strategy is null with predicate_known=true.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
533	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	21	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	—	●●■■■■■ ■■■ 0.20	No DB candidates exist for the predicate, and the search strategy is null with predicate_known=true.
loan application	REQUIRES_NUMBER_OF_FAMILY_MEMBERS	Not specified in the given context	Correct	factually_consistent	—	SEMANANTIC	—	●●■■■■■ ■■■ 0.20	Object indicates absence of information, which is treated as CORRECT per exception rules.
300	HAS_DESCRIPTION	smallest credit score	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	Predicate is not known in the schema, and no DB candidates exist for verification.
loan	HAS_MINIMUM_AMOUNT_USD	1000	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	Predicate is not known in the schema, and no DB candidates exist for verification.
loan	HAS_MAXIMUM_AMOUNT_USD	10000500	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	Predicate is not known in the schema, and no DB candidates exist for verification.

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.